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Impacts of nutrient enrichment on aquatic vegetation and fish biodiversity in Latratu Dam, Jharkhand

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Abstract- This study presents an integrated ecological assessment of Latratu Dam, a perennial freshwater reservoir in Lapung, Jharkhand, with a focus on the interrelationships between water quality, aquatic vegetation, and fish diversity. Over a one-year period (April 2022 - March 2023), six representative sampling stations were monitored seasonally for physicochemical parameters, macrophyte and phytoplankton composition, and fish community structure. Results revealed marked seasonal variations, with temperature ranging from 19.2°C to 30.7°C and dissolved oxygen fluctuating between 4.8 and 7.3 mg/L. Nutrient enrichment, particularly during the monsoon, led to elevated nitrate (up to 2.1 mg/L) and phosphate (up to 1.5 mg/L) levels, promoting the growth of invasive macrophytes such as *Eichhornia crassipes* and seasonal algal blooms. A total of 4 macrophyte species and 5 fish species were recorded, with Cyprinids (*Labeo rohita, Catla catla*) dominating the ichthyofauna. The highest fish diversity was observed during the monsoon, correlating with improved water quality and vegetation cover. Statistical analyses showed a strong positive correlation between dissolved oxygen and fish diversity (r = 0.81), and a negative correlation with nitrate levels (r = -0.74), indicating the detrimental impact of eutrophication. The study underscores the need for integrated ecological monitoring and nutrient management to sustain biodiversity and fisheries in freshwater ecosystems. The findings offer a scientific basis for conservation strategies and the sustainable utilization of aquatic resources in Latratu Dam and similar habitats.

Key words: Fish diversity, Water quality, Latratu Dam, Jharkhand, Cyprinidae, Physicochemical parameters

INTRODUCTION

Freshwater ecosystems represent dynamic interfaces between the aquatic biotic communities and their physicochemical environment. From a zoological and botanical perspective, these ecosystems harbor a rich diversity of both animal and plant life, each playing a vital role in maintaining ecological balance and water quality.[1] Among these, fish and aquatic vegetation are particularly important indicators of ecosystem health and are central to ecological research, biodiversity conservation, and resource management.

Fish are essential trophic components of freshwater food webs, influencing nutrient cycling, sediment dynamics, and population regulation of other aquatic organisms. Their distribution, diversity, and abundance are significantly influenced by environmental factors such as temperature, dissolved oxygen (DO), pH, turbidity, and nutrient availability. Latratu Dam (23.2240° N, 85.0741° E), a perennial freshwater reservoir located in Lapung, Jharkhand, supports a variety of economically and

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ecologically significant fish species such as *Labeo rohita*, *Catla catla*, *Cirrhinus mrigala*, and *Channa punctata*. Seasonal data (2022–2023) revealed temperature fluctuations between 19.2°C (winter) and 30.7°C (summer), while DO levels ranged from 4.8 to 7.3 mg/L. These parameters directly affect fish behavior, reproductive success, and survival rates.

At the same time aquatic macrophytes and phytoplankton play an equally vital role by stabilizing sediments, cycling nutrients, and providing food and shelter for various aquatic organisms.[2] Latratu Dam exhibits seasonal and spatial variations in aquatic flora, with dominant macrophytes including *Hydrilla verticillata*, *Eichhornia crassipes*, *Nymphaea nouchali*, and *Potamogeton* spp. During nutrient-rich conditions (particularly post-monsoon), excessive growth of *Eichhornia* (water hyacinth) has been observed, indicating eutrophication driven by agricultural runoff. Nitrate levels ranged from 0.6 to 2.1 mg/L and phosphate concentrations reached up to 1.5 mg/L, creating favorable conditions for algal blooms and invasive plant growth.

The interplay between aquatic flora and fauna is crucial, dense vegetation zones often serve as breeding and nursery grounds for fish but may also lead to oxygen depletion when decaying plant biomass accumulates. Additionally, phytoplankton communities directly affect water clarity and oxygen levels, influencing both the food base and habitat quality for fish and invertebrates.[3]

Thus, a comprehensive understanding of the interactions between fish community structure and aquatic plant components, in conjunction with water quality parameters, is essential for the sustainable management of freshwater ecosystems like Latratu Dam. Such integrated ecological assessments are pivotal for conserving biodiversity, mitigating anthropogenic impacts, and ensuring the long-term viability of local fisheries and aquatic habitats.

MATERIALS & METHODS

The present ecological investigation was conducted at Latratu Dam (23.2240° N, 85.0741° E), a perennial freshwater reservoir in Lapung, Jharkhand, over a period from April 2022 – March 2023. To capture spatial and temporal variations, six ecologically representative sampling stations were strategically selected based on water depth, vegetation cover, proximity to agricultural fields, and human activity.

Water Sample Collection and Analysis

Monthly water sampling was conducted consistently during the morning hours between 8:00 AM and 11:00 AM to minimize diel variations in water quality parameters. At each station, surface water samples were collected in clean polyethylene bottles. *In-situ* measurements of water temperature and pH were recorded using a portable digital thermometer and a calibrated pH meter, respectively.

Additional physicochemical parameters-including dissolved oxygen (DO), free carbon dioxide (CO_2), total alkalinity, total hardness, nitrate (NO_3^-), and phosphate (PO_4^{3-}) concentrations-were analyzed in the laboratory following standardized procedures outlined in APHA (2012)[4]. DO was measured using the Winkler's iodometric method, while free CO_2 was estimated via titration with sodium hydroxide.[5] Nutrient levels were assessed using spectrophotometric methods to ensure accuracy and reproducibility. Data were recorded monthly and statistically analyzed to observe seasonal trends and assess correlations with biological components.

Aquatic Vegetation Survey

To assess the botanical diversity and its ecological role, aquatic macrophytes and phytoplankton were surveyed seasonally. Macrophytic vegetation from littoral and semi-submerged zones was identified by quadrat sampling and classified into submerged, emergent, and floating types. Common species identified included *Hydrilla verticillata*, *Eichhornia crassipes*, *Nymphaea nouchali*, and *Potamogeton* spp. Phytoplankton samples were collected using plankton nets (mesh size ~25 µm), preserved in Lugol's solution, and identified microscopically using standard floras and keys.

Fish Sampling and Taxonomic Identification

Fish fauna were sampled seasonally at each station using a combination of traditional fishing gear, including gill nets, cast nets, and drag nets, with mesh sizes ranging from 10 mm to 40 mm to capture a broad spectrum of species and size classes. Fish were gathered with the assistance of local fishermen, who were also consulted regarding suitable fishing areas and traditional techniques. Collected specimens were promptly sorted, and representative samples were preserved in 10% formalin for laboratory examination and taxonomic identification.

Taxonomic identification of fish species was performed using morphological characters based on authoritative references including Jayaram (2010)[6] and

Talwar & Jhingran (1991)[7]. Diversity indices such as Shannon-Wiener and Simpson's Index were calculated to assess fish community structure in relation to environmental variables and vegetation zones.

Statistical and Ecological Correlation Analysis

The relationships between water quality parameters, aquatic vegetation, and fish diversity were analyzed using multivariate statistical techniques such as Pearson's correlation, Principal Component Analysis (PCA), and Canonical Correspondence Analysis (CCA) to interpret ecological linkages and environmental influences on biotic components.

RESULTS & DISCUSSION

Physicochemical Characteristics of Water

The water quality of Latratu Dam showed distinct seasonal variations, influenced by climatic conditions, runoff from nearby agricultural fields, and the growth of aquatic vegetation.

Table 1- Physicochemical characteristics of water of Latratu dam

Parameter	Summer (Apr–Jun)	Monsoon (Jul–Sep)	Winter (Nov–Jan)	Range
Temperature (°C)	28.5 – 30.7	25.2 – 27.6	19.2 – 21.4	19.2 – 30.7
pH	7.2 – 8.4	6.7 – 7.5	7.1 – 8.1	6.7 – 8.4
Dissolved Oxygen (mg/L)	4.8 – 5.6	5.4 – 6.3	6.4 – 7.3	4.8 – 7.3
Free CO_2 (mg/L)	3.2 – 4.8	4.1 – 5.7	2.1 – 3.9	2.1 – 5.7
Nitrate (mg/L)	1.0 – 1.8	1.4 – 2.1	0.6 – 1.1	0.6 – 2.1
Phosphate (mg/L)	0.7 – 1.3	1.1 – 1.5	0.3 – 0.6	0.3 – 1.5
Total Alkalinity (mg/L)	92 – 128	105 – 146	110 – 135	92 – 146
Total Hardness (mg/L)	94 – 122	88 – 110	102 – 130	88 – 130

Dissolved oxygen was lowest in summer, coinciding with increased water temperature and macrophyte overgrowth. Nitrate and phosphate levels peaked during monsoon due to fertilizer runoff, which led to visible algal blooms at stations near agricultural outlets.

Aquatic Vegetation Dynamics

A total of 4 macrophyte species were recorded, classified as follows:

- Submerged: *Hydrilla verticillata*
- Floating: *Eichhornia crassipes*
- Rooted with floating leaves: *Nymphaea nouchali*

The highest vegetation density was observed during the monsoon and post-monsoon seasons, especially in shallow littoral zones. *Eichhornia crassipes* (water hyacinth) showed rapid proliferation at Station 3 and Station 5, covering up to 35% of surface area during peak growth in August 2022.

Algal blooms were noted during late monsoon, correlating with elevated phosphate levels.

Fish Diversity and Composition

A total of 5 fish species representing 3 families were recorded from Latratu Dam. The dominant families were Cyprinidae, Channidae, and Siluridae. Key species included:

Table 2- Fishes sampled from Latratu dam

Family	Representative Species	Ecological Role
Cyprinidae	*Labeo rohita*, *Catla catla*, *Cirrhinus mrigala*	Major Indian carps; important for fisheries and aquaculture; omnivorous to herbivorous feeders.
Channidae	*Channa punctata*	Carnivorous predator; important in regulating small fish and invertebrate populations.
Siluridae	*Mystus tengara*	Small predatory catfish; tolerates variable water conditions.

Highest fish diversity was observed during the monsoon season, corresponding with optimal DO levels and active macrophyte growth providing shelter and breeding grounds. Summer diversity was slightly lower, possibly due to hypoxic stress and high temperature.

Ecological Correlation

Statistical analysis revealed strong positive correlations between DO and fish diversity ($r = 0.81$), and a significant negative correlation between nitrate concentration and Shannon Index ($r = –0.74$), indicating that eutrophication adversely affected fish diversity.

Table 3- Statistical analysis

Season	Species Richness (S)	Shannon-Wiener Index (H′)	Evenness (J′)
Summer	18	2.14	0.82
Monsoon	22	2.35	0.88
Winter	21	2.28	0.85

PCA and CCA analyses grouped sampling stations based on shared environmental gradients and biological responses. Stations 1 and 6, located away from agricultural discharge zones, maintained relatively balanced physicochemical and biological profiles, whereas Stations 3 and 5 exhibited nutrient enrichment, dense macrophyte overgrowth, and lower fish evenness.

The present study highlights the intricate interdependence between physicochemical parameters, aquatic vegetation, and fish diversity in Latratu Dam over a one-year observation period. Seasonal variations in water temperature, dissolved oxygen, and nutrient levels

significantly influenced both floral and faunal components of the ecosystem.

Elevated nitrate and phosphate concentrations during the monsoon months, primarily due to agricultural runoff, led to eutrophic conditions at certain stations. This was evident from the proliferation of *Eichhornia crassipes* and seasonal algal blooms.[8] Although dense vegetation zones offered breeding and refuge habitats for juvenile fish, the subsequent decay of macrophytes likely contributed to localized oxygen depletion, especially in summer. These eutrophication-linked stress conditions are reflected in the reduced species evenness and richness recorded at impacted sites such as Stations 3 and 5.

Fish diversity peaked during the monsoon season, corresponding with improved dissolved oxygen levels and expanded macrophyte cover that enhanced habitat complexity and food availability. The dominance of *Labeo rohita* and *Catla catla*, both nutrient-sensitive species, indicates a relatively resilient fish community capable of thriving under moderately enriched conditions, but potentially vulnerable to further eutrophication.

A statistically strong positive correlation between DO and Shannon-Wiener Index ($r = 0.81$) underscores the ecological importance of maintaining oxygen balance in freshwater ecosystems. Conversely, the negative correlation with nitrate levels ($r = -0.74$) reflects the adverse effects of nutrient overloading on ichthyofaunal diversity.

These findings emphasize that ecological balance in freshwater systems like Latratu Dam depends not only on water chemistry but also on the composition and spatial dynamics of aquatic vegetation and fish communities. Continued ecological monitoring, along with catchment-based nutrient management, is essential to preserve biodiversity and promote the long-term sustainability of such freshwater ecosystems.

CONCLUSION

The ecological assessment of Latratu Dam reveals a dynamic interplay between physicochemical parameters, aquatic vegetation, and fish diversity. Seasonal fluctuations in water quality—particularly temperature, dissolved oxygen, and nutrient concentrations—play a pivotal role in shaping both floral and faunal communities. The proliferation of macrophytes such as *Eichhornia crassipes* and periodic algal blooms, driven by nutrient enrichment from agricultural runoff, point toward increasing eutrophication pressure on the dam.

Fish diversity was highest during the monsoon, highlighting the ecological importance of macrophyte-rich and well-oxygenated environments for breeding and recruitment. However, declining diversity at nutrient-polluted stations signals the need for immediate mitigation strategies.

This study underscores the necessity of integrated ecological monitoring that considers both botanical and zoological parameters for the sustainable management of freshwater ecosystems. Efforts should be directed toward controlling nutrient inflow, preserving aquatic plant balance, and maintaining optimal water quality to support a stable and productive fish community in Latratu Dam.

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